📊 Portfolio Optimization Project

Data Integrity & Foundation

Data Source

- **Period:** Jan 2023 – Dec 2024 (2 Full Calendar Years)
- **Source:** Yahoo Finance API (yfinance)
- **Stocks:** 15 Indian stocks + Nifty 50

Data Cleaning Process

- **5-Day Hard Filter:** Pruned stocks missing >5 consecutive trading days
- **Forward Filling:** Minor gaps filled with last known mark-to-market price
- **Result:** Pristine dataset ready for analysis

16	730	11,680	100%
Total Instruments	Trading Days	Data Points	Data Integrity

The Risk-Return Landscape

Scatter Plot Analysis

Annualized Volatility vs. Return visualization reveals sector clustering and efficient frontier.

Stock	Return %	Vol %	Sharpe	Sector
HCLTECH	42.71%	21.37%	1.69	IT
SUNPHARMA	40.23%	17.37%	1.94	Pharma
DIVISLAB	36.61%	26.26%	1.15	Pharma
KOTAKBANK	-1.01%	20.47%	-0.37	Banking

Key Insights

- **IT Sector (TCS/INFY/HCLTECH):** Clusters in high-return, moderate-risk zone
- **Pharma Sector:** Shows defensive behavior with strong risk-adjusted returns
- **Efficient Frontier:** Top-left stocks = "Alpha generators" (high return, low risk)
- **Mispriced Asset:** KOTAKBANK is an outlier – high risk for negative returns

Alpha vs. Beta: Best & Worst Performers

🏆 Top 3 Performers (by Sharpe Ratio)

Rank	Stock	Return	Volatility	Sharpe
1	SUNPHARMA.NS	40.14%	17.37%	1.94
2	HCLTECH.NS	42.71%	21.37%	1.69
3	DRREDDY.NS	29.85%	19.45%	1.20

⚠️ Bottom 3 Performers (by Sharpe Ratio)

Rank	Stock	Return	Volatility	Sharpe
1	KOTAKBANK.NS	-1.01%	20.47%	-0.37
2	HDFCBANK.NS	5.81%	19.80%	-0.03
3	AXISBANK.NS	6.62%	21.90%	0.01

Strategic Takeaway

The Winners: Dominated due to high "Return-to-Risk" efficiency – these are true Alpha generators.

The Losers: These weren't just low-return stocks; they were "Risk Traps" – high volatility with negative or flat gains.

→ This justifies why Portfolio B excludes the bottom 3 entirely.

Technical Momentum: SMA Signals

Golden Cross Signals (Bullish)

Stocks with SMA-50 above SMA-200 – Go signals

Stock	SMA-50	SMA-200	Signal Date
HCLTECH	1793.46	1560.20	2024-07-22
SUNPHARMA	1789.58	1662.53	No crossover
INFY	1824.44	1650.15	2024-07-12
TCS	3988.15	3898.83	No crossover

Death Cross Signals (Bearish)

Stocks with SMA-50 below SMA-200 – Blacklisted from allocation

Stock	SMA-50	SMA-200	Signal Date
KOTAKBANK	351.79	353.39	2024-12-26
CIPLA	1486.84	1493.14	2024-12-24
AXISBANK	1143.20	1160.50	2024-12-12
DRREDDY	1264.83	1268.54	2024-12-13

Analysis: Trend Physics

→ **Golden Crosses:** Our "Go" signals – strong bullish momentum confirmed

→ **Death Crosses:** Any stock showing Death Cross as of Dec 31 was **immediately blacklisted** from recommended allocation to avoid capital erosion

The Master Strategy: Portfolio Comparison

📉 Portfolio A (Equal Weight)

- **Strategy:** Buy-and-Hope
- **Allocation:** Equal 1/15 for all 15 stocks
- **Total Investment:** ₹10,00,000
- **Expected Return:** ~18.5%
- **Beta:** 0.80
- **10% Nifty Drop:** -7.97% loss

📈 Portfolio B (Optimized)

- **Strategy:** Engineered (Max-Sharpe + SMA Filter)
- **Stocks:** Only 6 selected (40% of original)
- **Top Holdings:**
  - SUNPHARMA: 41.70% (₹4,16,968)
  - HCLTECH: 28.64% (₹2,86,361)
  - ICICIBANK: 10.93% (₹1,09,270)
  - DIVISLAB: 11.51% (₹1,15,147)
- **Beta:** 0.63 (21% lower risk)
- **10% Nifty Drop:** Only -6.30% loss

+21%	21%	15%	0%
Sharpe Improvement	Beta Reduction	Crash Buffer	Upside Sacrifice

ML Strategy: Relative Outperformance Classifier

Model Methodology

Instead of risky price forecasting, we built a **Supervised Classification Model**.

Goal: Binary Classification (Yes/No)
Predicts whether a stock will **outperform the Nifty 50 benchmark** over the next 30-day window based on technical features.

Top 3 Feature Drivers

1. **Moving Average Divergence (SMA 50/200):** Measures trend strength and momentum persistence.
2. **Daily Return Volatility:** Assesses risk-adjusted stability.
3. **Beta vs. Benchmark:** Measures sensitivity to systemic market shocks.

Financial Interpretation

Our ML analysis confirms that **structural momentum and technical trend alignment** are statistically significant predictors of outperformance in the NSE. The model effectively filters stocks with high 'Alpha' potential from those merely tracking market 'Beta,' allowing us to refine Portfolio B's selection criteria beyond standard historical averages.

Automation & Monitoring: n8n Workflow

Live Monitoring System

📊 Data Fetch → 🔍 Signal Check → ⚡ Alert Trigger → 📧 Email/Telegram

Trigger Conditions

- **Price Shock:** >2% price movement
- **SMA Crossover:** Golden/Death Cross detection
- **Portfolio Drift:** Allocation deviation >5%

The Value Proposition

"This workflow automates the 'Human-in-the-loop' requirement by monitoring Portfolio B for 2% price shocks and SMA crossovers, ensuring the Fund Manager can react in **minutes, not days**."